Quiz 10

1. Consider the double pendulum shown in Fig. 1. The pendulum system only moves in the x-y plane.

   (a) What is the minimum number of generalized coordinates necessary to fully describe the motion of the double pendulum? (In other words, how many degrees of freedom does this system have?) 2

   (b) In terms of the minimum number of generalized coordinates, write the x- and y-components of the position of m_1, and then write the x- and y-components of the position of m_2.

   $\vec{r}_1 = l_1 \sin\phi_1 \hat{x} - l_1 \cos\phi_1 \hat{y}$

   $\vec{r}_2 = (l_1 \sin\phi_1 + l_2 \sin\phi_2)\hat{x} - (l_1 \cos\phi_1 + l_2 \cos\phi_2)\hat{y}$

   (c) Write the x- and y-components of the velocities of m_1 and m_2 in terms of the generalized coordinates.

   $\vec{v}_1 = l_1 \cos\phi_1 \dot{\phi}_1 \hat{x} + l_2 \sin\phi_1 \dot{\phi}_1 \hat{y}$

   $\vec{v}_2 = (l_1 \cos\phi_1 \dot{\phi}_1 + l_2 \cos\phi_2 \dot{\phi}_2)\hat{x} + (l_1 \sin\phi_1 \dot{\phi}_1 + l_2 \sin\phi_2 \dot{\phi}_2)\hat{y}$

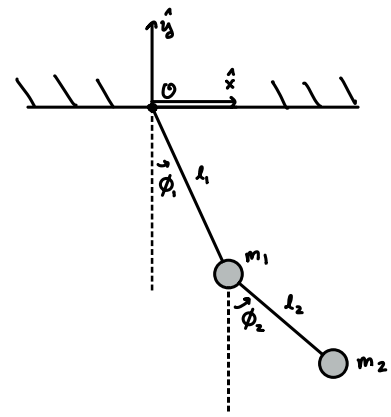

Figure 1: A double pendulum. A string of fixed length l_1 is attached to the ceiling and a mass m_1. A string of fixed length l_2 is attached to a mass m_1 and a mass m_2.

2. Consider a system with n degrees of freedom and n generalized coordinates $q_1, q_2, ..., q_n$

   (a) The path this system follows is determined by the Lagrange equations when the following conditions are met (circle those that are true):

   i. There are no constraint forces present
   ii. All nonconstraint forces are conservative
   iii. All nonconstraint forces are not conservative

   (b) Assuming the right conditions in the previous part holds, how many Lagrange equations determine the path the system follows? n

   (c) Write one of the Lagrange equations that describes the motion of this system.

$$\frac{\partial \mathcal{L}}{\partial q_1} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_1}$$